Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 3 W device runs for 5 s. Find transferred energy.

Marzano check - circle: ① ② ③

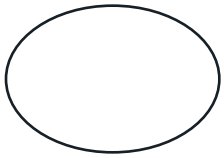
02 Worked example: explain the model

Reference: $\Phi = BA$; $|emf_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and mark the changing field through the loop. Contrast it with a stationary case.

Label field direction, loop area and what changes.



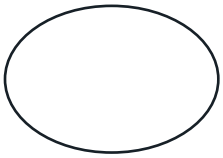
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

03 Read and interpret the model

Compare a magnet held still beside a fixed coil with a magnet moving into the coil. Which can produce a changing flux?

Label field direction, loop area and what changes.



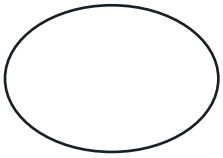
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

04 Apply the physics

A lab group pushes a magnet into a fixed coil and records a positive meter deflection. Draw the approach and withdrawal motions. Predict the sign when the magnet is pulled back out along the same path without changing connections.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep charge separation fixed and multiply one charge magnitude by 4. Compare electric-force magnitude. Label charges and both interaction forces.



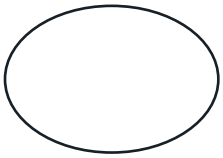
Charge A Charge B

Marzano check - circle: 1 2 3

07 Exit check

Name two ways to increase the magnitude of induced emf in a simple magnet-and-coil trial.

Label field direction, loop area and what changes.



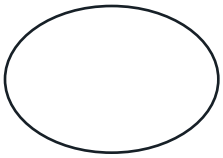
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Predict how faster approach and more turns can change induced voltage. State which comparisons must keep the other conditions fixed.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3